Polymath3 now active

Terence Tao · 30 September 2010

https://terrytao.wordpress.com/2010/09/30/polymath3-now-active/

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Gil Kalai has officially started the Polymath3 project (Polynomial Hirsch conjecture) with a research thread at his blog.

The original aim of this project is to prove the polynomial Hirsch conjecture, which is a conjecture in the combinatorial geometry of polytopes. However, there is a reduction due to Eisenbrand, Hahnle, Razborov, and Rothvoss that would deduce this conjecture from a purely combinatorial conjecture, which can be stated as follows.

> **Combinatorial polynomial Hirsch conjecture.** Let $\mathcal{F}_1, \ldots, \mathcal{F}_t$ be non-empty collections of subsets of $\{1, \ldots, n\}$ with the following properties:
>
> 1. (Disjointness) $\mathcal{F}_i \cap \mathcal{F}_j = \emptyset$ for every $i \neq j$.
> 2. (Connectedness) If $1 \leq i < j < k \leq t$ and $A \in \mathcal{F}_i, B \in \mathcal{F}_k$, there exists $C \in \mathcal{F}_j$ such that $A \cap B \subset C$.
>
> Then t is of polynomial size in n (i.e. $t = O(n^{O(1)})$).

For instance, when n=3, one can obtain such a family with t=6, e.g.

$$\mathcal{F}_1 = \{\emptyset\}, \mathcal{F}_2 = \{\{1\}\}, \mathcal{F}_3 = \{\{1,2\}\}, \mathcal{F}_4 = \{\{1,2,3\}\},$$
$$\mathcal{F}_5 = \{\{2,3\}\}, \mathcal{F}_6 = \{\{3\}\};$$

one can show that this is the best possible value of t for this choice of n. The best possible value of t for n=4 is still not worked out; it is between 8 and 11.

One appealing thing about this problem is that there is a simple elementary argument that gives the bound $t \leq n^{\log_2 n + 1}$ for all $n \geq 2$; and so in some sense one is "only a logarithm away" from proving the conjecture. Anyway, the project is just starting, and does not require any particularly specialised background, so anyone who may be interested in this problem one may want to take a look at the research thread.

Comments (7)

 **Américo Tavares** · 1 October 2010

Typo: missing ‘latex’ in $A \cap B \subset C$, $A \cap B \subset C$. *[Corrected, thanks – T.]*

 **Scott Carnahan** · 1 October 2010

I think the Disjointness axiom needs an empty set. *[Corrected, thanks – T.]*

 **Johan** · 2 October 2010

There is one bracket too much in the definition of F_5. *[Not sure I see this... – T.]*

 **Anonymous** · 2 October 2010

I think you’ve got one too many bracket also, should be F_5 = {{2,3},{3}}, rather than {{2,3}},{{3}}
[Ah, I see the problem now: F_6 had run off beyond the column width of the blog and so was not visible. Fixed now – T.]

 **David Hansen** · 2 October 2010

Should “polytops” be “polytopes”?
[Corrected, thanks – T.]

 **Eric** · 12 October 2010

Why is
{\mathcal F}_1 = \{ \emptyset\}, {\mathcal F}_2 = \{ \{1\}\}, {\mathcal F}_3 = \{\{1,2\}\}, {\mathcal F}_4 = \{\{1,2,3\}\}, {\mathcal F}_5 = \{\{2,3\}\}, {\mathcal F}_6 = \{\{3\}\};
optimal?
Why cant I shove in {{2}} to get
{\mathcal F}_1 = \{ \emptyset\}, {\mathcal F}_2 = \{ \{1\}\}, {\mathcal F}_3 = \{\{2\}\}, {\mathcal F}_4 = \{\{1,2\}\}, {\mathcal F}_5 = \{\{1,2,3\}\}, {\mathcal F}_6 = \{\{2,3\}\}, {\mathcal F}_7 = \{\{3\}\};
(I may be making a silly error, but I cannot see it)

 **Eric** · 12 October 2010

Nevermind!! Please ignore my post, I see that {1}\cap {1,2,3}={1} not subset of {2}. I should think about things more before posting next time.